1. Given a set of lines of text of arbitrary length, design and implement a function that will left and right justify the text in a way that avoids splitting words and leaves paragraphs indented. An attempt should also be made to distribute the additional blanks as evenly as possible in the justified line.

2. Write a program that will search a line of given text for a particular pattern or substring. Should the pattern be found it is to be replaced by another given pattern.

3. Consider the following subsets of tokens:

Keyword = {if, for, while, int, float, char}
Operator = {+, -, *, /, =,<,>}
Constant = (any unsigned integer between 1 and 999)
Identifier = (any string with 8 characters or less, the first character must be an alphabet, the rest are any combination of alphabets/digits/underscores)
Symbols = { , ( ) ; ?}

Now, given a program statement in C, classify the tokens used in the statement on the basis of the above-mentioned subset of tokens using transition diagrams. The program should generate appropriate error message on detection of invalid tokens.